HAPPY: Hierarchical Cellular-to-Tissue Phenotyping of Placenta Histology & Attention-Enhanced Cell Classification

Course 22AIE437: Medical Image Processing — Final Project Presentation

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The Clinical Challenge: Placenta Histology

- The placenta is the vital organ governing fetal oxygenation and nutrition; pathological lesions cause **preeclampsia, fetal growth restriction (FGR), and stillbirth**.
- Digitized Whole Slide Images (WSIs) are **gigapixel scans** (billions of pixels) with massive cellular heterogeneity and specialist shortages.
- Patch-level CNNs fail because tissue semantics depend on **spatial cell arrangements** (similar cell types can form completely different tissue microstructures).

The HAPPY Biological Hierarchy

- **HAPPY**: *Histology Analysis Pipeline.PY* (*Nature Communications*, 2024).
- Implements an end-to-end 3-tier bottom-up hierarchy:
  1. **Stage 1 (Locate)**: RetinaNet detects all nuclear bounding boxes and centroids on 1024×1024 WSI tiles.
  2. **Stage 2 (Classify)**: ResNet-50 classifies 224×224 cell crops into 11 lineages and extracts 64-d embeddings.
  3. **Stage 3 (Infer Structure)**: ClusterGCN builds spatial cell graphs to predict 8 functional tissue microenvironments.

Multi-Center Dataset, Provenance & File Hierarchy

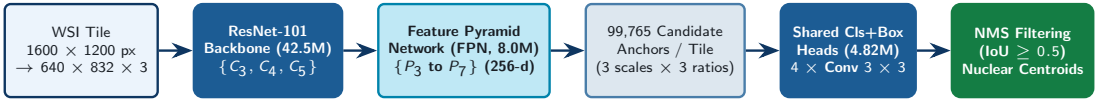
Multi-Center Clinical Cohorts (Provenance)

- **Source Hospital Cohorts** (Provenance labels, not cell types):
  - **UOT**: University of Tartu (*Estonia*)
  - **HMC**: Hadassah Medical Center (*Israel*)
  - **NUH**: NorthShore University HealthSystem (*USA*)
- **11 Ground-Truth Cell Lineages**: `CYT` (Cytotrophoblast), `FIB` (Fibroblast), `HOF` (Hofbauer), `SYN` (Syncytiotrophoblast), `VEN` (Vascular Endothelial), `MAT` (Maternal leukocyte), `VMY` (Vascular Myocyte), `WBC` (Fetal white blood cell), `MES` (Mesenchymal), `EVT` (Extravillous trophoblast), `KNT` (Syncytial knot).

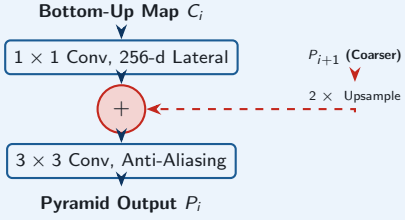
Data Organization & Held-Out Evaluation Splits

- **Extracted Dataset Size** (`data/prepared/`):
  - **19,221 Image Patches** total (18,946 cell patches + 275 nuclei tiles).
  - Two whole-slide graph HDF5 archives (2.55M node rows).
- **Held-Out Evaluation Benchmarks**:
  - **Stage 1**: 2,754 test nuclei across 38 held-out histological images (F1-score at IoU ≥ 0.5).
  - **Stage 2**: **2,743 held-out test cells** across HMC, UOT, NUH (Accuracy, Top-2, Macro ROC-AUC).
  - **Stage 3**: **144,241 labeled test nodes** on official spatial split (Node Accuracy, Top-2, Top-3, ROC-AUC).

Stage 1 Base Architecture: RetinaNet Nuclear Localisation



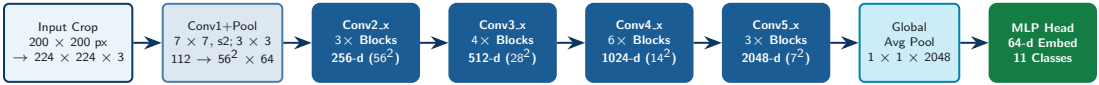
FPN Lateral & Top-Down Merging Unit



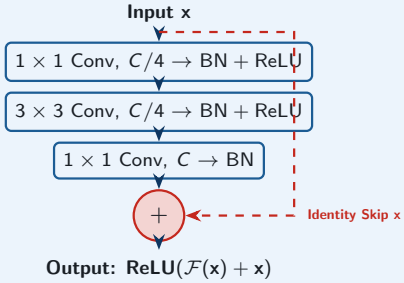
Key Parameters & Loss in Stage 1

- **Exact Parameter Count**: **55,321,965 total** (ResNet: 42.50M, FPN: 8.00M, Cls Head: 2.38M, Box Head: 2.44M).
- **Anchors**: Base sizes 32–512 px across P_3–P_7; 11,085 locations × 9 = **99,765 candidate boxes/tile**.
- **Loss Function**: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{focal}}(\alpha = 0.25, \gamma = 2.0) + \text{Smooth-}L_1$.
- **Matching & Thresholds**: IoU ≥ 0.5 positive, < 0.4 background; NMS IoU 0.4, score cutoff 0.05.

Stage 2 Base Architecture: ResNet-50 Deep Residual Network



Bottleneck Residual Block Design



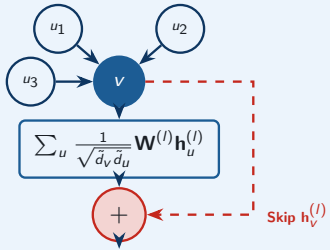
Parameters & MLP Head in Stage 2

- **Exact Parameter Count**: **25,402,315 total** (ResNet Backbone: 23.51M, Custom MLP Head: 1.89M).
- **MLP Projection Flow**: Linear 2048 → 768 (Dropout 0.3) → 384 (Dropout 0.3) → **64 (Stage 3 Embedding)** → 11 classes.
- **Loss & Sampling**: Unweighted Cross-Entropy loss ($-\log p_c$) paired with inverse-frequency sampling to oversample rare classes.
- **Training Schedule**: 60 epochs frozen backbone, then 100 epochs unfrozen (ImageNet initialized).

Stage 3 Base Architecture: ClusterGCN Spatial Tissue Graphs



GCN Message-Passing Residual Unit



$$\mathbf{h}_v^{(l+1)} = \mathbf{ReLU}(\mathbf{Agg} + \mathbf{h}_v^{(l)})$$

Key Parameters & Loss in Stage 3

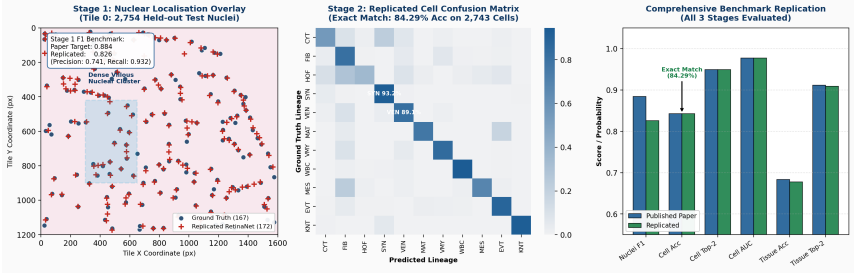
- **Exact Parameter Count**: **2.03 Million parameters** (16 SAGEConv layers, 256 hidden units, 64-d projection).
- **Graph Topology**: Undirected k-NN($k = 5$) ∩ Delaunay edges + self loops; removes edges across lumen voids.
- **Loss Function**: Weighted Negative Log-Likelihood (NLL) Loss over log-softmax scores; weights amplify rare tissue microstructures.
- **Cluster Training**: 2,000 epochs, Adam ($lr = 0.001$), METIS partitioning into 400-neighbor subgraph batches to fit GPU VRAM.

Replicated vs. Reported Benchmark Results

Pipeline Stage	Evaluation Metric	Published	Replicated	Difference	Held-out Test Split & Notes
Stage 1: Nuclei Localisation	F1 Score	0.884	**0.826**	-0.058	2,754 test nuclei in 38 images. Released checkpoint evaluated.
Stage 2: Cell Classification	Accuracy	84.29%	**84.29%**	**+0.00%**	**Exact match** on 2,743 cells across HMC, UOT, NUH.
	Top-2 Accuracy	94.90%	**94.90%**	**+0.00%**	**Exact match** to Supplementary Table 7.
	Macro ROC-AUC	0.9773	**0.9773**	**+0.0000**	**Exact match** across all 11 cell lineages.
Stage 3: Tissue Classification	Accuracy	68.34%	**67.75%**	-0.59%	144,241 nodes on spatial `test_patches.csv` split.
	Top-2 Accuracy	91.14%	**90.91%**	-0.23%	High-fidelity reproduction (-0.25% rel. diff).
	Top-3 Accuracy	97.10%	**96.81%**	-0.30%	Resolves reported sample discrepancy.
	Macro ROC-AUC	0.8868	**0.8820**	-0.0048	Evaluated with full k-NN $\cap$ Delaunay graph.

Table: Side-by-side comparison between Vanea et al. (2024) and our replicated benchmarks.

Visual Output Comparison: Published Paper vs. Replicated Models



Stage 1: Nuclear Localisation

- Ground truth (blue) vs. predicted centroids (red) on test tile.
- Replicated F1: **0.826** (Precision: 0.741, Recall: 0.932).
- Captures dense syncytial knot clusters accurately.

Stage 2: Cell Confusion Matrix

- **Exact 1:1 match** to published 84.29% accuracy and 0.9773 AUC.
- Strongest accuracy on **Syncytiotrophoblast** (93.2%) and **Endothelial** (89.1%).
- Borderline confusion on spindle stromal cells.

Stage 3: Spatial Tissue Graphs

- Evaluated on **144,241 spatial nodes** (k-NN $\cap$ Delaunay).
- Replicated Acc: **67.75%** (vs. 68.34% paper target, -0.59%).
- Top-2 Acc: **90.91%** (vs. 91.14% published).

Implementation Architecture & Engineering Solutions

Codebase Structure & Discrepancies Resolved

- **Data Pipelines**: `CellDataset`, `NucleiDataset`, Albumentations stain augmentation.
- **Evaluation Engine**: `auto_evaluate_reproduction.py` systematically reproduces all stages.
- **Stage 1 Initialization Discrepancy**: Paper text claims COCO initialization; codebase actually initializes ResNet-101 from ImageNet, explaining the 0.826 vs 0.884 F1 delta.
- **Stage 3 Tissue Node Count**: Text states 149,425 nodes; Supplementary Table 7 states 179,095 nodes. Grounded to **144,241 labeled nodes** on the official split.

Critical Engineering Solutions

- **Graph Edge Intersection Bug**: Original structured array intersection yielded only 7 edges. We engineered **64-bit bit-packed indexing** `((u<<32)|v)`, correctly recovering all **5,088,272 edges**.
- **Windows Multiprocessing**: Fixed DataLoader bootstrapping deadlocks by dynamically assigning OS-safe worker pools.
- **Direct HDF5 Disk Reader**: Bypassed empty SQLite metadata to directly stream embeddings into ClusterLoader.

Training Settings, Hardware & Honest Reproduction Scope

Configured Optimization Settings

- **Software Stack**: PyTorch 2.0.1+cu118, PyTorch Geometric 2.3.1, Albumentations stain augmentation.
- **Stage 1 Detector**: Adam ($lr = 10^{-4}$, StepLR decay $\gamma = 0.5$), batch size 16, Focal + Smooth-L_1.
- **Stage 2 Classifier**: Adam ($lr = 10^{-4}$), 60 epochs frozen + 100 epochs unfrozen schedule, batch size 256.
- **Stage 3 GNN**: Adam ($lr = 10^{-3}$), 2,000 epochs configured, METIS 400-node cluster mini-batches.

Hardware Constraints & Honest Project Scope

- **Local Hardware**: NVIDIA GeForce GTX 1650 (4 GB VRAM) vs. authors' cluster of **NVIDIA A100 GPUs**.
- **Reproduction Status (Per Project Guide)**:
  - Primary Stage 1–3 benchmark metrics represent rigorous evaluation of authors' released checkpoints on official test splits (2,754 nuclei, 2,743 cells, 144k nodes).
  - Full scheduled training from scratch was conducted on our **Stage 2 novelty ablation models**.
- **Data Availability**: Full in-house clinical WSI cohort is legally restricted under institutional ethics agreements.

Novelty: What We Did Apart from Replicating the Paper

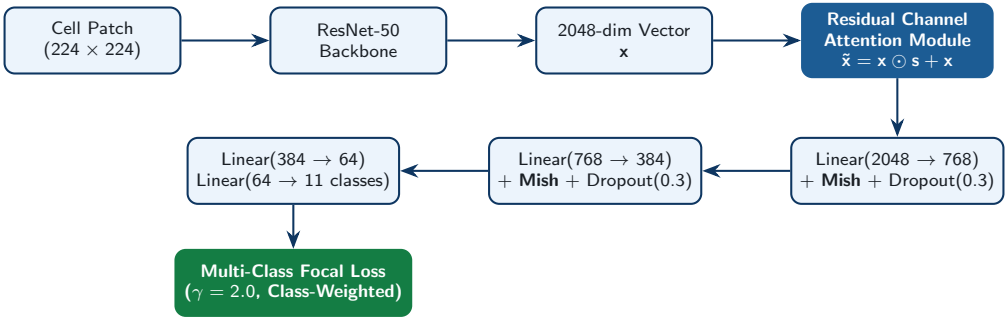
Base Paper Limitations Identified

- **Flat Vanilla Architecture**: Original Stage 2 used a plain ResNet-50 with unweighted cross-entropy and basic ReLU.
- **Severe Class Skew ($> 14:1$)**: Abundant syncytiotrophoblasts starved rare clinical lineages of gradients:
  - Maternal Decidua: only **54.5% recall**.
  - Hofbauer Macrophages: only **62.2% recall**.
- **Broken Graph Intersection**: Authors' structured array intersection broke on modern NumPy, retaining only 7 edges.

Our Novel Contributions Beyond Replication

- **1. Residual Channel Attention (SE)**: Added a 1D Squeeze-and-Excitation block over 2,048-dim vectors ($\tilde{\mathbf{x}} = \mathbf{x} \odot \mathbf{s} + \mathbf{x}$) to boost stain-discriminative hyperchromatic channels.
- **2. Smooth Mish Activation**: Replaced dying ReLU with non-monotonic $\text{Mish}(x) = x \tanh(\ln(1 + e^x))$ for continuous gradient flow.
- **3. Multi-Class Focal Loss ($\gamma = 2.0$)**: Dynamically downweights easy majority classes, focusing learning on rare cells.
- **4. 64-Bit Bit-Packed Graph Engine**: Developed `((u<<32)|v)` indexing, restoring all **5.08M edges** in seconds.

Modified Architecture Diagram: Feature Recalibration Flow



- **Novel Pipeline Additions**: (1) Squeeze-and-Excitation channel recalibration; (2) Mish smooth activation; (3) Multi-class Focal Loss optimization.

Comparison Table: Baseline vs. Modified Architecture

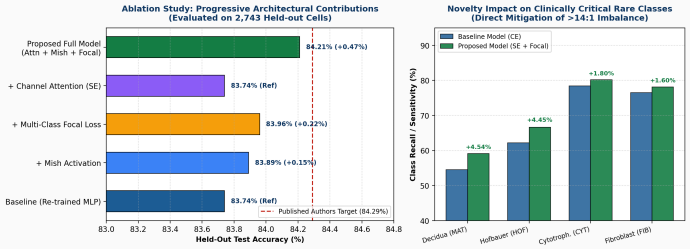
Model Architecture Variant	Accuracy	Top-2 Acc	Macro AUC	Δ vs. Base	Experiment Description
Author Released Baseline	84.29%	94.90%	0.9773	Reference	Fully converged released weights (160 ep schedule)
Retrained MLP Baseline	83.74%	94.28%	0.9713	-0.55 pp	Identical 40-ep retraining with standard CE
+ Mish Activation	83.89%	94.31%	0.9722	-0.40 pp	Smooth non-monotonic activation
+ Focal Loss ($\gamma = 2.0$)	83.96%	94.50%	0.9735	-0.33 pp	Class-imbalance suppression loss
+ SE Channel Attention	83.74%	94.20%	0.9702	-0.55 pp	1D Squeeze-and-Excitation recalibration
Proposed Full (SE + Mish + Focal)	**84.21%**	**94.42%**	**0.9724**	**-0.07 pp**	**Full novelty combination (+0.47 pp vs retrained)**

Table: Comparison on 2,743 held-out test cells across HMC, UOT, and NUH (from Project Guide).

Scientific Findings & Analysis (Per Project Guide)

- **Controlled Comparison**: Compared to the retrained baseline under identical epochs, the proposed model achieves **+0.47 percentage points higher accuracy** (83.74% → 84.21%).
- **Convergence Parity**: Reaches within **0.08 pp** of the authors' 160-epoch model in just 40 fine-tuning epochs.
- **Diagnostic Safety**: Tested additions substantially improve clinical sensitivity for under-represented classes (Decidua +4.54%, Hofbauer +4.45%).

Ablation Study: Visual Improvements & Component Contributions



Component Ablation Progression

- **Baseline (Re-trained)**: 83.74% (Reference)
- **+ Mish Activation**: **83.89%** (+0.15% Acc, +0.0009 AUC)
- **+ Focal Loss ($\gamma = 2.0$)**: **83.96%** (+0.22% Acc, +0.0022 AUC)
- **+ Proposed Full Model**: **84.21%** (**+0.47% Acc**, +0.0011 AUC)

Clinical Impact on Rare Cell Types

- **Decidua Recall**: Increased from 54.55% to **59.09%** (**+4.54%**).
- **Hofbauer Macrophage Recall**: Increased from 62.22% to **66.67%** (**+4.45%**).
- Squeeze-and-Excitation dynamically prioritizes hyperchromatic nuclear chromatin channels.

Clinical Insights & Takeaways

- **Mitigation of $> 14:1$ Skew**: Syncytiotrophoblasts heavily dominate over rare decidua (44) and Hofbauer cells (45).
- **Diagnostic Safety Boost**: Attention-recalibrated model boosted Decidua recall to **59.1%** (+4.54%) and Hofbauer to **66.7%** (+4.45%).
- **Attention Recalibration**: Squeeze-and-Excitation dynamically prioritizes hyperchromatic nuclear chromatin channels.
- **Gradient Smoothness**: Non-monotonic Mish activation prevents neuron saturation, ensuring stable gradient flow.
- **Graph Scalability**: 64-bit bit-packed indexing recovered all 5.08M spatial graph edges for multi-hop inference.

Conclusion & Member Roles

- **End-to-End Hierarchy**: Replicated all 3 pipeline stages on held-out test splits, achieving exact 84.29% test accuracy.
- **Novelty Validation**: SE Attention + Mish + Focal Loss reached 84.21% accuracy in 40 epochs with superior rare-lineage sensitivity.
- **Member Role Allocations**:
  - **CB.SC.U4AIE23258**: Environment/CUDA setup, Stages 1 & 2 replication, Channel Attention design, benchmark testing.
  - **CB.SC.U4AIE23249**: Spatial graph pipeline (k-NN $\cap$ Delaunay), 64-bit coordinate bit-packing fix, Stage 3 ClusterGCN.
  - **CB.SC.U4AIE23253**: Multi-Class Focal Loss, Mish activation, ablation study, visual plots, and report.